0580_s22_ms_42_Q11

Question	Answer	Marks	Partial Marks
11(a)(i)	4.455 to 4.456… [= 4.46]	**2**	**M1** for [r =] $\frac{28}{2\pi}$ oe
11(a)(ii)	1250 or 1247 to 1249.9…	**2**	**M1** for $20 \times \pi \times 4.46^2$ oe
11(a)(iii)	66[.0] or 65.95 to 66.02	**3**	**M2** for [tan] = $\frac{20}{2 \times 4.46}$ oe or **B1** for identifying angle *ANB* on cylinder **not on rectangle**
11(b)	11.8 or 11.82 to 11.83	**5**	**M2** for [r =] $\sqrt[3]{\frac{310 \times 3}{2\pi}}$ oe or [h =] $\sqrt[3]{\frac{310 \times 3 \times 4}{\pi}}$ oe or **M1** for $310 = \frac{1}{3}\pi \times r^2 \times 2r$ or $310 = \frac{1}{3}\pi\left(\frac{h}{2}\right)^2 h$ **M2** for $\sqrt{(their\ r)^2 + (2 \times their\ r)^2}$ oe or **M1** for [l^2 =]$(their\ r)^2 + (2 \times their\ r)^2$ oe